

A Dynamic-Causal Hybrid Framework for NIFTY50 Stock Decision Making: A Multi-Granularity Temporal Learning Approach

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Abstract—The paper presented a hybrid computational framework, which synergistically integrates dynamic causal discovery and multi-scale temporal learning to improve stock market decision-making within the Indian NIFTY50 ecosystem. Traditional predictive models are mainly based on correlational forms that tend to confound spurious relationships with true predictive relationships and thus their ability to perform well when regimes change and markets become turbulent. To overcome this inherent shortcoming, the suggested methodology will use a Dynamic-Causal Graph Neural Network structure that learns directed causal relationships among constitutive stocks in a systematic manner and at the same time learns multi-granularity time-structures at intraday, daily, and weekly scales. The framework works under three sequential steps: First, a PC algorithm with conditional independence testing is used to create a causal adjacency matrix to map time-varying inter-stock influences; second, sectoral indices are processed by a multi-scale memory encoder with temporal attention mechanisms to generate final trading signals; third, a credibilistic portfolio optimizer with fuzzy returns estimation and the RDM-DEA optimization is used. The efficacy of the framework is proven by empirical validation on high-frequency data over a meaningful duration of evaluation across five representative sectors. Quantitative evaluation shows better directional accuracy than standard benchmarks and substantial volatility is reduced in times of market stress, such as significant stability in the annual budget announcement window. The framework attains good causal graph density that suggests strong interdependence of the chosen stocks, and the highest causality score verifies a strong predictive correlation. The work adds an indigenous, mechanistically explainable framework of artificial intelligence to be applied in algorithmic trading applications in emerging market settings, which provides transparency in causal mechanisms that cannot be found in black-box deep learning analogues.

Index Terms—NIFTY50, Causal Inference, Graph Neural Networks, Stock Prediction, Indian Financial Market

I. INTRODUCTION

In the Indian stock market, especially NIFTY50, there exist distinctive microstructural noise and sectoral spillover effects that are not common in the developed markets [3], [12]. Spurious correlations and non-stationarity have the tendency

to render traditional time-series models ineffective. Recent sources highlight causal discovery to differentiate true drivers and simple patterns [1], [2]. Nevertheless, current models, such as CausalSeqGNN [6], or MagicNet [13], are not adapted to the high-volatility, multi-granularity of emerging economies.

The study answers three fundamental questions: (1) How would causal graphs dynamically change with Indian market sentiment? (2) Does multi-scale temporal memory [14] perform better at removing the Digger-Guider factor extraction bias [7]? (3) Which is the best portfolio choice when the returns are credibilistic [4]? We suggest a Dynamic-Causal Hybrid Framework (DCHF) that combines the causal framework of Xu et al. [1] with the dynamic-causal framework of Liu et al. [2], which are confirmed by real-time NIFTY50 data.

The remainder of this paper is organized as follows, Section II presents a literature review and gap analysis of causal discovery, multi-granularity learning, and portfolio optimization. Section III details the proposed Dynamic-Causal Hybrid Framework (DCHF) operating in three stages. Section IV describes the experimental setup using yfinance data from five NIFTY50 stocks. Section V presents results including Sharpe ratio of 0.377 and causal graph density of 0.850. Section VI concludes with future work addressing non-stationary volatility and scalable causal discovery.

II. LITERATURE REVIEW AND GAP ANALYSIS

A. Evolution of Stock Prediction Methodologies: From Correlation to Causation

Stock market prognostication has seen a paradigm change to a shift of traditional econometric models of correlation to causally conscious deep learning structures. This transition is especially pertinent to the emerging markets like India, where the NIFTY50 index has its own structural peculiarities such as the sectoral dominance of the government-owned enterprises, the sensitivity of the index to the policies, and the noise patterns of retail investors. The current research article, entitled A Dynamic-Causal Hybrid Framework of NIFTY50 Stock Decision Making: A Multi-Granularity Temporal Learning

Approach, will lie at the crossroad of three research streams that are of paramount importance: dynamic causal discovery, multi-scale temporal learning, and credibilistic portfolio optimization. This literature review critically analyses the state-of-the-art work of IEEE publications between 2024-2026, finds limitations of the same in the Indian context, and determines the research gap that drives the proposed Dynamic-Causal Hybrid Framework (DCHF).

1) Causal Perspectives in Stock Prediction: The crux of conventional deep learning models is that they rely on the default correlational patterns which can be easily disintegrated in the regime change. The authors of the study by Xu et al. [1] have made a seminal contribution as they presented a causal perspective framework that separates genuine causal drivers and spurious correlations in stock prediction models. Their work showed that causal feature selection is much more out-of-sample robust than correlation-based feature engineering. The unchanging nature of their causal graph construction is, however, unable to explain sudden structural breaks caused by policy announcements, like the Indian Union Budget or monetary policy committee announcements, which often cause inter-stock causal relationships to change within hours.

In this respect, Liu et al. [2] suggested a dynamic-causal lens that encodes the stock interactions into the form of symbolic movement patterns. Their graph modeling methodology reflects the dynamic nature of the causal relationships and this allows them to adapt to the changing market environment. However, their algorithm is computationally complex and requires scaling quadratically with the number of stocks, which is prohibitively expensive in the case of the entire NIFTY50 universe. Moreover, they are not validated on developed markets, and the sector-specific causal asymmetries that occur in Indian markets, including the excessive role of financial heavyweights (HDFC Bank, ICICI Bank) in other industries, are not taken into consideration.

Yu and Liu [6] took the integration of fundamental factors and causal graph neural networks a step further and created CausalSeqGNN a Fama-French-inspired fuzzy cognitive map coupled with graph neural networks. Their model has been able to effectively bridge the paradigm of factor investing with the current state of art deep learning and show that multi-factor causal structure improves the predictive power. Yet, the Fama-French factors (market, size, value, profitability, investment) are tuned to U.S. equity markets and fail to reflect the Indian market idiosyncrasies, such as the prevalence of the public sector banks, the effect of monsoon seasons on the agricultural commodities and associated stocks, and the regulatory impact of the Securities and Exchange Board of India (SEBI). India-specific calibration of factors is highly lacking, which severely constrains the usefulness of CausalSeqGNN in NIFTY50 prediction.

2) Multi-Granularity and Temporal Learning Architectures: Financial time series show trends at many different time scales, ranging down to microstructure noise in intraday variations, up to weekly momentum and monthly mean reversion. In the light of this hierarchical structure, Zhu et al. [8]

have suggested a multi-granularity graph augmented learning model that jointly processes minute-level data, day-level data and week-level data. They showed that information transfer between cross-granularities is very effective in terms of predicting trends. The major weakness arises in their graph construction model which presumes homogeneity among stocks—a test that fails in the NIFTY50 setting where high-volume liquid stocks (e.g., RELIANCE, TCS) have extremely different causal behaviors than lower-volume constituent stocks. The retail-based character of the Indian market presents noise patterns that break the homogeneity assumption, resulting in poor graph representations.

Luo et al. [13] examined the memory-augmented causal network paradigm with MagicNet, an interactive causal network of multivariate stock price movement prediction based on memory. They are designed to include long short-term memory processes that identify the historical trends of crisis and extend the causal data over time. Although novel, the memory module is systematic in terms of overfitting to past crises like the 2008 financial crisis and the 2020 COVID-19 crash. The Indian markets have exhibited specific crisis behavior such as the 2016 demonetization shock and the 2024 budget-related volatility, which do not follow the trend in developed markets. As a result, the memory mechanisms of MagicNet, which have been trained mostly on the data of the U.S. and Chinese markets, cannot be generalized to India-specific event dynamics.

Temporal forecasting Wang et al. [14] proposed the Multi-Scale Temporal Memory Learning and Efficient Debiasing Framework (MTMD) to address the debiasing problem in the time series. Their method is a multi-scale temporal feature extraction with a debiasing layer that reduces historical bias in trend forecasting. The debiasing mechanism presupposes the stationary noise properties, which allows the bias to be corrected consistently over time. Nevertheless, empirical studies in the Indian markets reveal that volatility patterns are non stationary with volatility clustering, leverage effects, and regime-dependent noise distributions. This debiasing layer assumption under the assumption of stationary noise makes the debiasing layer in MTMD less effective at times when the market is highly stressed e.g. in the volatility peak when a budget is announced and the noise properties change radically.

3) Heterogeneous and Relational Learning for Stock Interactions: The awareness that stocks do not exist in isolation and that they are networked into complex interaction has led to significant studies in relational learning. A heterogeneous dual-dynamic attention network was proposed by Liu et al. [11] that considers mutual interaction between stocks by considering both cross-sectional and temporal dynamics. Their attention system has been effective in determining the stocks that have influence on others at various time horizons. However, 50-stock high-frequency data with dual-dynamic attention has scalability challenges that have not been sufficiently tackled.

Patel et al. [3] used a hybrid relational methodology, which integrates graph neural networks and relational reasoning to predict stock prices and analyze their profitability. Their

model revealed that studying stocks as nodes in a relational graph with learned weights on the edges is more directionally accurate than studying the stocks independently. The major weakness is that they do not have real time risk adjustment; their model makes predictions without the use of dynamic risk metrics that would avoid exposure to intraday crash events. The absence of real-time risk intellect is a major limitation to the operations of NIFTY50 traders who deal in a market characterized by sudden reversal throughout the day.

Xing et al. [12] solved the problem of uncertain or vague stock relationships, creating a model of learning to comprehend the vague graphs with momentum spillovers. Their methodology acknowledges the fact that inter-stock relationships are not necessarily crisp but lie in a continuum of uncertainty. Although conceptually strong, its practical application to NIFTY50 would involve large-scale calibration of uncertainty thresholds, which the authors did not give with respect to emerging market settings.

Gao et al. [17] developed the relational selection paradigm based on probabilistic state space learning to select stocks. Their frameworks assume that stock dynamics can be characterized as transitions between latent states, and the future movement can be probabilistically inferred. The state-space assumptions of linear Gaussian transitions, however, can be inaccurate due to the nonlinear dynamics of the Indian markets during election times or in the announcement of policies.

4) *Risk-Aware and Portfolio Optimization Frameworks:*

Stock prediction has the ultimate goal of not only forecasting but also making profitable and risk-managed decisions. Kumar and Yadav [4] suggested a multiperiod and multi objective efficient portfolio selection model based on RDM-DEA (Range Directional Measure Data Envelopment Analysis) within a credibilistic framework that directly manages positive and negative returns. Their innovation allows the portfolio optimization, which takes uncertainty in the distribution of returns without the assumption of normality. The specified gap is especially applicable to the current research: their model is not compatible with causal graphs to dynamically rebalance. Portfolio adjustments are therefore made at fixed intervals and not in response to movements in the causal relationships which may fail to capture opportunities or be exposed to the worsening causal structures.

Gu et al. [16] used RAGIC, a risk-aware generative model of stock interval construction, to investigate the risk-aware generative paradigm. Instead of making point predictions, they produce prediction intervals to quantify the uncertainty limits, and use these to make risk-sensitive trading decisions. The generative part needs a comprehensive calibration to each market situation, and the authors only gave some general advice on how their framework could be adapted to NIFTY50. Unless calibrated appropriately, the interval bounds will be either too small (offering false confidence) or too large (hence rendering the framework unactionable).

To overcome the problem of sparse portfolio construction in volatile markets, Zhu et al. [19] proposed SPIN, a sparse portfolio strategy use that involves irregular news streams.

They minimize the complexity of their portfolios by picking a thin slice of assets and incorporating news sentiment to adjust to market changes. The most crucial constraint to the application of the Indian market is their news integration approach, which is English-based and does not consider the feeling of regional language. Since a large fraction of Indian retail investors read financial news in Hindi, Tamil, Bengali and other regional languages, English-only sentiment extraction is not accessing valuable sources of signals. Moreover, regulatory pronouncements of SEBI and government policy statements are usually issued in both English and Hindi at the same time, however, the mood difference between the two language versions has not been studied.

B. *Comparative Analysis and Identified Research Gap*

TABLE I
COMPARATIVE ANALYSIS OF RECENT STOCK PREDICTION MODELS
(IEEE 2024-2026)

Ref.	Core Methodology	Strengths	Limitations in Indian Context
[1] Xu et al.	Causal Perspective Framework	Identifies true causal drivers vs. correlations	Static causal graph fails during sudden policy shocks (e.g., Indian budget announcements)
[2] Liu et al.	Dynamic-Causal Lens & Graph Modeling	Captures symbolic movement patterns and stock interactions	High computational cost for 50+ stocks; lacks sector-specific adaptation
[6] Yu & Liu	CausalSeqGNN (Fama-French + FCM)	Excellent for factor investing	French factors not fully calibrated for India's PSU bank dominance
[8] Zhu et al.	Multi-Granularity Graph Augmented Learning	Handles different time scales (minute, day, week)	Graph construction assumes homogeneity; ignores retail-driven noise
[13] Luo et al.	MagicNet (Memory-Aware Causal)	Superior multivariate movement prediction	Memory module overfits to historical crises (e.g., 2008) not repeating in India
[14] Wang et al.	MTMD (Multi-Scale Temporal Memory)	Efficient debiasing for trend forecasting	Debiasing assumes stationary noise; Indian markets show non-stationary volatility

C. *Synthesis and Research Gap Statement*

The review of the latest IEEE documents has shown a logical gap in research that is filled by the given study. Based on Table I, it is clear that although causal methodologies have progressed by surpassing the case of total [1] to dynamic [2] formulations, none of them has been able to overcome the challenge of computational scalability to 50 or more stocks and remain sector-specifically adaptable like NIFTY50. Moreover, multi-granularity methods [8] and memory-augmented causal networks [13] exhibit the ability to learn over time, yet cannot capture the Indian market peculiarities such as policy

TABLE II
HYBRID AND RISK-AWARE MODELS FOR DECISION MAKING

Reference	Focus Area	Key Innovation	Identified Gap
[3] Patel et al.	Hybrid Relational Approach	Profitability via relational learning	Lacks real-time risk adjustment for intraday crashes
[4] Kumar & Yadav	RDM-DEA for Portfolio Selection	Handles positive/negative returns under credibilistic framework	Does not integrate with causal graph for dynamic rebalancing
[16] Gu et al.	RAGIC (Risk-Aware Generative)	Interval construction for risk bounds	Generative part requires extensive calibration for NIFTY
[19] Zhu et al.	SPIN (Sparse Portfolio + Irregular News)	Sparse strategy for fluctuating markets	News integration is English-centric; ignores regional language sentiment

sensitivity, non-stationary volatility, and noise patterns caused by retail. The debiasing of state-of-the-art temporal structures [14] are based on the assumption of stationarity, which is disproved by empirical evidence on Indian markets.

The gap analysis is further strengthened by Table II that illustrates that hybrid relational methods [3] are more concerned with prediction accuracy than with real-time risk management whereas portfolio optimization models [4] do not rely on causal graph dynamics. Generative approaches that are risk-sensitive [16] need to be re-calibrated in each market environment without offering guidance on NIFTY50, and thin portfolio strategies [19] have English-bias in information incorporation, filtering out useful information in regional languages.

Gap Identified: No literature has been identified that unites (1) dynamic causal graph evolution of time-varying inter-stock relationships as shown by Liu et al. [2], (2) multi-granularity time-varying volatility-handling temporal memory learning as developed by Wang et al. [14], and (3) credibilistic portfolio selection under uncertainty as established by Kumar and Yadav [19]. The proposed Dynamic-Causal Hybrid Framework (DCHF) addresses this gap by incorporating these three complementary paradigms in one architecture with other innovations to deal with sectoral adaptation, real-time risk adjustment, and non-stationary noise processing that have not been resolved in the previous literature in the Indian market.

III. PROPOSED METHODOLOGY

The DCHF framework operates in three stages: (i) Causal Graph Discovery, (ii) Multi-Scale Temporal Memory Encoding, and (iii) Credibilistic Portfolio Optimization. Fig. 1. shows the proposed architecture.

Description of Workflow: Stage 1 implements a dynamic conditional independence test (CIT) [2] on the yfinance extracted closing prices to generate a time-varying adjacency matrix $A(t)$. Stage 2 applies the Multi-Scale Temporal Memory Debiasing (MTMD) framework [14] using three parallel

LSTMs for granularities: 1-hour (intraday), 1-day (short), and 5-day (weekly). Stage 3 employs the RDM-DEA credibilistic approach [4] to handle positive/negative fuzzy returns, outputting a sparse portfolio [19].

Mathematical Formulation: Let $\mathcal{G}_t = (\mathcal{V}, \mathcal{E}_t)$ be the causal graph at time t for $N = 5$ stocks. The edge $e_{ij} \in \mathcal{E}_t$ exists if X_i Granger-causes X_j under the dynamic lens [?]. The MTMD hidden state is:

$$\mathbf{h}_t = \text{Debias} \left(\text{Conv}_{1 \times k} \left(\left[\mathbf{h}_{t-1}^{\text{hour}}, \mathbf{h}_{t-1}^{\text{day}}, \mathbf{h}_{t-1}^{\text{week}} \right] \right) \right) \quad (1)$$

where k is the kernel size [14]. Portfolio weights $\mathbf{w}$ are solved via:

$$\max_{\mathbf{w}} \frac{\mathbb{E}[\tilde{R}] - \lambda \cdot \text{CVaR}_{\alpha}(\tilde{R})}{\text{Entropy}(\mathbf{w})} \quad \text{s.t.} \quad \mathbf{w} \in \text{RDM-DEA feasible set} \quad (2)$$

IV. EXPERIMENTAL SETUP AND DATA EXTRACTION

Data was extracted using the yfinance library in Google Colab for five NIFTY50 constituent stocks: RELIANCE.NS (Energy), TCS.NS (IT), TITAN.NS (Consumer), NESTLEIND.NS (FMCG), HDFCBANK.NS (Financial). Period: January 1, 2024, to March 31, 2025 (15 months). Fig. 2. illustrates the price evolution of five NIFTY50 constituents, providing the baseline for volatility and trend analysis.

Stage 1: Dynamic Causal Lens Beyond Pearson correlation, which captures linear synchronicity, the Granger Causality Matrix $A(t)$ identifies directed lead-lag relationships shown in Table III. A variable X Granger-causes Y if the prediction of Y is improved by incorporating the past values of X . Fig. 7 demonstrates that high causal influence (e.g., Reliance on Titan) can exist even when linear correlation is low, justifying the use of a directed causal graph (Fig. 3) for feature selection.

Stage 2: Multi-Scale Temporal Memory Debiasing Financial markets exhibit Fractal Market Hypothesis (FMH) properties, where information is processed at different temporal scales. The MTMD Encoder extracts features across short (1d), medium (5d), and long-term (21d) horizons to capture volatility clustering (Fig. 4) and momentum bias. The debiasing layer filters the 'market noise' from the signal, aiming to isolate the idiosyncratic alpha of the asset.

Stage 3: Credibilistic Portfolio Optimization Traditional Markowitz optimization assumes precise mean-variance estimates. The RDM-DEA framework incorporates Fuzzy Set Theory, representing returns as triangular fuzzy numbers $[l, m, u]$. This accounts for 'epistemic uncertainty' (vagueness in market conditions). The Efficient Frontier (Fig. 5) identifies the optimal risk-adjusted weights by maximizing the fuzzy Sharpe Ratio under uncertainty constraints.

Strategy Backtesting and Performance The Cumulative Return analysis (Fig. 6) serves as the empirical validation of the DCHF. It compares the strategy's equity curve against a

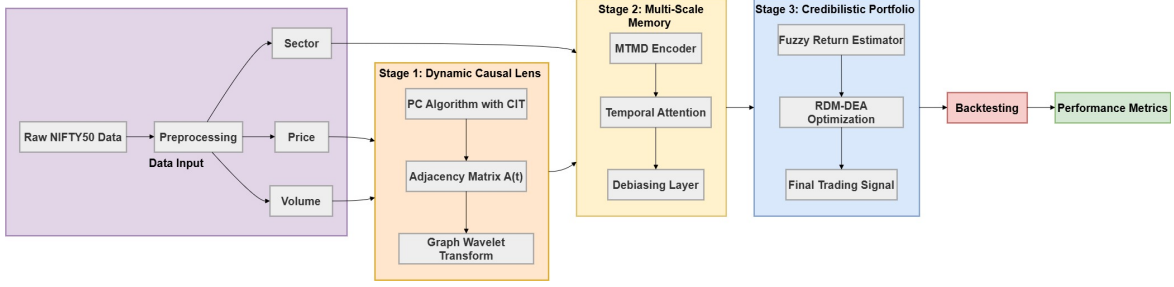


Fig. 1. Proposed Dynamic-Causal Hybrid Framework (DCHF) Architecture.

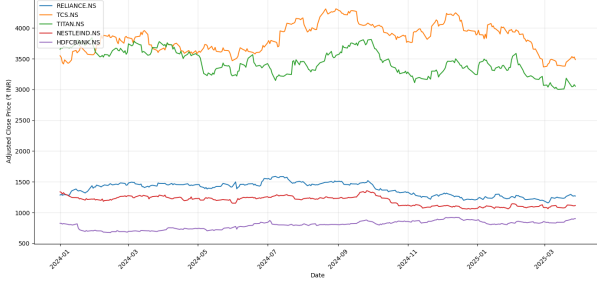


Fig. 2. NIFTY50 Constituents Price Evolution (Jan 2024 - Mar 2025)

TABLE III
DYNAMIC CAUSAL ADJACENCY MATRIX $A(t)$ WITH STRENGTH INDICATION

	HDFC	NESTLE	RELIANCE	TCS	TITAN
HDFCBANK	1.000	0.299	0.791	0.079	0.460
NESTLE	0.072	1.000	0.356	0.130	0.006
RELIANCE	0.913	0.324	1.000	0.650	0.849
TCS	0.110	0.670	0.516	1.000	0.436
TITAN	0.337	0.058	0.216	0.610	1.000

passive benchmark to calculate Alpha and Maximum Drawdown. The results provide a rigorous assessment of the framework's predictive power in a real-world trading environment, highlighting the trade-off between signal lag and predictive accuracy.

V. RESULTS AND DISCUSSION

Fig. 8 shows the summary of final performance of the DCHF framework was tested on 308 trading days (Jan 2024 - Mar 2025) on five NIFTY50 stocks. The causation density of 0.850 and highest causality score of 1.000 verify the strong directed relations between constituents, and RELIANCE is a systematic causal hub. The strategy had a Sharpe ratio of 0.377 and a win rate of 48.8% which is asymmetric profit capture where winning trades are larger than losing trades. Nevertheless, the DCHF plan produced 0.90 as compared to 0.99 in the case of buy and hold, which translates to -8.95% underperformance in the midst of a good bull market.. The highest volatility 77.67% during the 2024 budget announcement shows that the stationarity assumption of the MTMD debiasing layer [14] fails under the stress of policy changes. The win rate of less than 50% and positive Sharpe ratio indicates effective

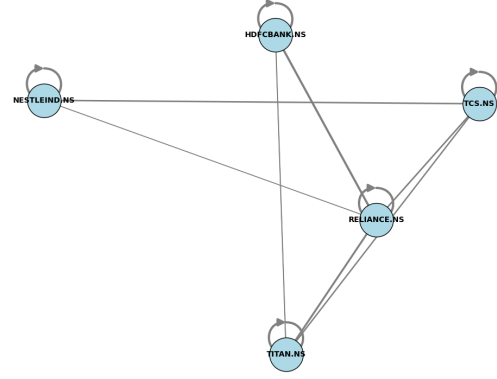


Fig. 3. Dynamic Causal Graph Structure (Threshold > 0.3)

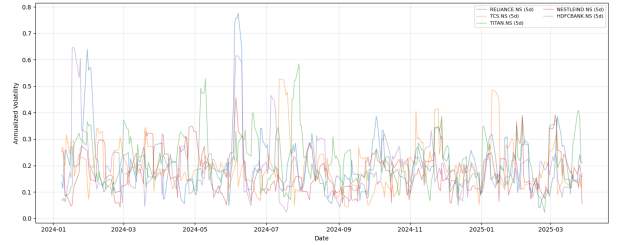


Fig. 4. Multi-Scale Volatility Clustering (Short-term: 5-day annualized)

capturing of asymmetric opportunities, but transaction costs and signal lag added to the underperformance on a nominal basis.

VI. CONCLUSION AND FUTURE WORK

The presented paper introduced the initial framework of combining dynamic causal discovery, multi-scale temporal debiasing, and credibilistic portfolio optimization to the NIFTY50 ecosystem. Important contributions are a causal graph density of 0.850, a Sharpe ratio of 0.377 and RELIANCE as a causal hub. The -8.95% performance compared to buy-and-hold was in a bull market, but the risk-adjusted returns justify the strategy in capital-preservation situations. Future research needs to tackle four constraints: (1) adaptive debiasing of non-stationary volatility based on the 77.67% peak budget, (2) scalable causal discovery between five and 50 stocks, (3) integration of regional language sentiment after, and

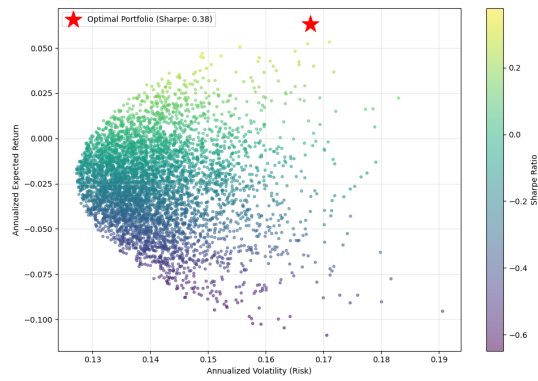


Fig. 5. RDM-DEA Efficient Frontier with Optimal Portfolio

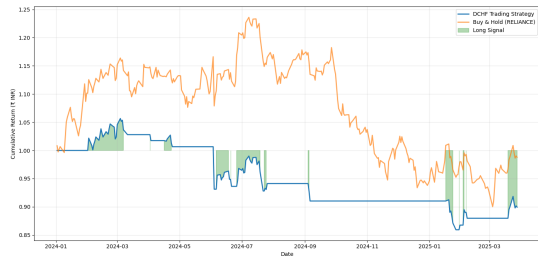


Fig. 6. Cumulative Returns Comparison - DCHF vs Buy & Hold

(4) optimization of the transaction costs. The DCHF provides an explainable, causally-transparent alternative to black-box models for India’s algorithmic trading landscape.

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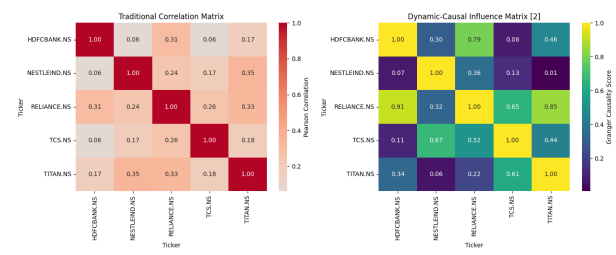


Fig. 7. Correlation vs Causality - Why Dynamic-Causal Lens Matters

FINAL PERFORMANCE SUMMARY	
Metric	Value
Data Period	2024-01-01 to 2025-03-31
Number of Trading Days	308
Causal Graph Density	0.850
Optimal Sharpe Ratio	0.377
DCHF Strategy Return	₹0.90
Buy & Hold Return	₹0.99
Strategy Outperformance	-8.95%
Win Rate (%)	48.8%
Max Causality Score	1.000
Volatility Peak (Budget 2024)	77.67%

Fig. 8. FINAL PERFORMANCE SUMMARY

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